1.32 Inverse Hyperbolic Functions

Worked Solutions

1)

$y = \sinh x$

Swap x and y:

$$x = \sinh y \Rightarrow x = \frac{e^y - e^{-y}}{2} \Rightarrow 2x = e^y - e^{-y}$$
$$\Rightarrow 2xe^y = e^{2y} - 1 \Rightarrow 0 = e^{2y} - 2xe^y - 1$$
$$\Rightarrow 0 = (e^y - x)^2 - x^2 - 1 \Rightarrow (e^y - x)^2 = x^2 + 1$$
$$\Rightarrow e^y - x = \pm\sqrt{x^2+1} \Rightarrow e^y = x \pm \sqrt{x^2+1}$$

Now $e^y > 0$ and $x - \sqrt{x^2+1} < 0$, so

$$e^y = x + \sqrt{x^2+1} \Rightarrow y = \ln(x + \sqrt{x^2+1})$$

$\therefore \operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$.

2)

$y = \cosh x$ is many-to-one, so we take $x > 0$ to restrict the domain to make it one-to-one.

Swap x and y:

$$x = \cosh y \text{ where } y > 0 \Rightarrow x = \frac{e^y + e^{-y}}{2}$$
$$\Rightarrow 2x = e^y + e^{-y} \Rightarrow 2xe^y = e^{2y} + 1 \Rightarrow 0 = e^{2y} - 2xe^y + 1$$
$$\Rightarrow 0 = (e^y - x)^2 - x^2 + 1 \Rightarrow (e^y - x)^2 = x^2 - 1$$
$$\Rightarrow e^y - x = \pm\sqrt{x^2-1} \Rightarrow e^y = x \pm \sqrt{x^2-1}$$
$$\Rightarrow y = \ln(x \pm \sqrt{x^2-1})$$

Now, $(x + \sqrt{x^2-1})(x - \sqrt{x^2-1}) \equiv x^2 - (x^2-1) \equiv 1$.

So $x - \sqrt{x^2-1} \equiv \frac{1}{x+\sqrt{x^2-1}} \equiv (x + \sqrt{x^2-1})^{-1}$.

Then $\ln(x - \sqrt{x^2-1}) \equiv \ln(x + \sqrt{x^2-1})^{-1} \equiv -\ln(x + \sqrt{x^2-1})$.

$$\Rightarrow y = \pm\ln(x + \sqrt{x^2-1})$$

As $y > 0$, we must have that $y = \ln(x + \sqrt{x^2-1})$.

$\therefore \operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$ where $x \geq 1$.

3)

$y = \tanh x$

Swap x and y:

$$x = \tanh y \Rightarrow x = \frac{e^{2y} - 1}{e^{2y} + 1} \Rightarrow x(e^{2y} + 1) = e^{2y} - 1$$
$$\Rightarrow xe^{2y} + x = e^{2y} - 1 \Rightarrow x + 1 = e^{2y} - xe^{2y}$$
$$\Rightarrow 1 + x = (1 - x)e^{2y} \Rightarrow e^{2y} = \frac{1+x}{1-x}$$
$$\Rightarrow 2y = \ln(\frac{1+x}{1-x}) \Rightarrow y = \frac{1}{2}\ln(\frac{1+x}{1-x})$$

$\therefore \operatorname{artanh} x = \frac{1}{2}\ln(\frac{1+x}{1-x})$ where $-1 < x < 1$.

4 a)

$\operatorname{arsinh} 1 = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2})$

4 b)

$\cosh^{-1} 2 = \ln(2 + \sqrt{2^2 - 1}) = \ln(2 + \sqrt{3})$

4 c)

$$\operatorname{artanh} \frac{1}{3} = \frac{1}{2} \ln(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}}) = \frac{1}{2} \ln 2$$

4 d)

$\sinh^{-1}(-4) = \ln(-4 + \sqrt{(-4)^2 + 1}) = \ln(-4 + \sqrt{17})$

4 e)

$\operatorname{arcosh} 9 = \ln(9 + \sqrt{9^2 - 1}) = \ln(9 + \sqrt{80}) = \ln(9 + 4\sqrt{5})$

4 f)

$$\tanh^{-1}(-\frac{1}{2}) = \frac{1}{2} \ln(\frac{1 + (-\frac{1}{2})}{1 - (-\frac{1}{2})}) = \frac{1}{2} \ln \frac{1}{3} = -\frac{1}{2} \ln 3$$

5 a)

$\sinh x = 0$

$$\Rightarrow x = \operatorname{arsinh} 0 = \ln(0 + \sqrt{0^2 + 1}) = \ln 1 = 0$$

5 b)

$\sinh x = 3$

$$\Rightarrow x = \operatorname{arsinh} 3 = \ln(3 + \sqrt{3^2 + 1}) = \ln(3 + \sqrt{10})$$

5 c)

$\sinh x = -6$

$$\Rightarrow x = \operatorname{arsinh}(-6) = \ln(-6 + \sqrt{(-6)^2 + 1}) = \ln(-6 + \sqrt{37})$$

6 a)

$\cosh x = 1$

$$\Rightarrow x = \pm \operatorname{arcosh} 1 = \pm \ln(1 + \sqrt{1^2 - 1}) = \pm \ln 1 = 0$$

6 b)

$\cosh x = 6$

$$\Rightarrow x = \pm \operatorname{arcosh} 6 = \pm \ln(6 + \sqrt{6^2 - 1}) = \pm \ln(6 + \sqrt{35})$$

6 c)

$\cosh x = \frac{3}{2}$

$$\Rightarrow x = \pm \operatorname{arcosh} \frac{3}{2} = \pm \ln(\frac{3}{2} + \sqrt{(\frac{3}{2})^2 - 1})$$

$$\Rightarrow x = \pm \ln(\frac{3}{2} + \sqrt{\frac{5}{4}}) = \pm \ln(\frac{3+\sqrt{5}}{2})$$

7 a)

$\tanh x = 0$

$$\Rightarrow x = \operatorname{artanh} 0 = \frac{1}{2} \ln(\frac{1+0}{1-0}) = \frac{1}{2} \ln 1 = 0$$

7 b)

$\tanh x = \frac{1}{4}$

$$\Rightarrow x = \operatorname{artanh} \frac{1}{4} = \frac{1}{2} \ln(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}) = \frac{1}{2} \ln \frac{5}{3}$$

7 c)

$\tanh x = -\frac{5}{6}$

$$\Rightarrow x = \operatorname{artanh}(-\frac{5}{6}) = \frac{1}{2} \ln(\frac{1+(-\frac{5}{6})}{1-(-\frac{5}{6})}) = \frac{1}{2} \ln \frac{1}{11} = -\frac{1}{2} \ln 11$$

8 a)

$\sinh 2x = 5$

$$\Rightarrow 2x = \operatorname{arsinh} 5 \Rightarrow 2x = \ln(5 + \sqrt{5^2+1}) \Rightarrow x = \frac{1}{2} \ln(5 + \sqrt{26})$$

8 b)

$\cosh 3x = 8$

$$\Rightarrow 3x = \pm \operatorname{arcosh} 8 = \pm \ln(8 + \sqrt{8^2 - 1}) \Rightarrow x = \pm \frac{1}{3} \ln(8 + \sqrt{63})$$

$$\Rightarrow x = \pm \frac{1}{3} \ln(8 + 3\sqrt{7})$$

8 c)

$5 \tanh(3x - 1) = 2$

$$\Rightarrow \tanh(3x-1) = \frac{2}{5} \Rightarrow 3x - 1 = \operatorname{artanh} \frac{2}{5}$$

$$\Rightarrow 3x - 1 = \frac{1}{2} \ln(\frac{1+\frac{2}{5}}{1-\frac{2}{5}}) = \frac{1}{2} \ln \frac{7}{3}$$

$$\Rightarrow x = \frac{1}{3}(\frac{1}{2} \ln \frac{7}{3} + 1) = \frac{1}{6} \ln \frac{7}{3} + \frac{1}{3}$$

8 d)

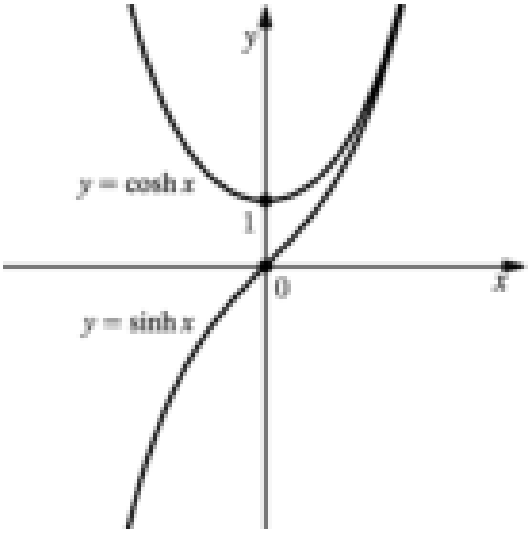
$3\sinh(2-\frac{1}{3}x)=1$

$$\Rightarrow \sinh(2-\frac{1}{3}x)=\frac{1}{3}\Rightarrow 2-\frac{1}{3}x=\operatorname{arsinh}\frac{1}{3}$$

$$\Rightarrow 2-\frac{1}{3}x=\ln(\frac{1}{3}+\sqrt{(\frac{1}{3})^2+1})=\ln(\frac{1}{3}+\sqrt{\frac{10}{9}})=\ln(\frac{1+\sqrt{10}}{3})$$

$$\Rightarrow x=3(2-\ln(\frac{1+\sqrt{10}}{3}))=6-3\ln(\frac{1+\sqrt{10}}{3})$$

9 a)



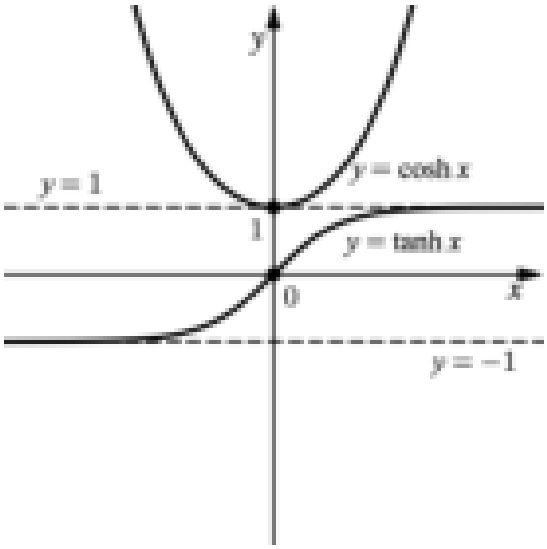
9 b)

$y=\sinh x$ and $y=\cosh x$ intersect if there are solutions to $\sinh x=\cosh x$.

$$\Rightarrow \frac{\sinh x}{\cosh x}=1\Rightarrow \tanh x=1\Rightarrow x=\operatorname{artanh}1=\frac{1}{2}\ln(\frac{1+1}{1-1})$$ which is undefined

So $y=\sinh x$ and $y=\cosh x$ do not intersect.

10 a)



10 b)

$y = \cosh x$ and $y = \tanh x$ intersect if there are solutions to $\cosh x = \tanh x$.

$$\Rightarrow \cosh x = \frac{\sinh x}{\cosh x} \Rightarrow \cosh^2 x = \sinh x$$

$$\Rightarrow 1 + \sinh^2 x = \sinh x \Rightarrow \sinh^2 x - \sinh x + 1 = 0$$

$$\Rightarrow (\sinh x - \frac{1}{2})^2 - (\frac{1}{2})^2 + 1 = 0 \Rightarrow (\sinh x - \frac{1}{2})^2 = -\frac{3}{4} \text{ which has no real solutions.}$$

So $y = \cosh x$ and $y = \tanh x$ do not intersect.

11 a)

$\sinh^2 x + \sinh x - 12 = 0$

$$\Rightarrow (\sinh x + 4)(\sinh x - 3) = 0 \Rightarrow \sinh x = -4 \text{ or } \sinh x = 3$$

If $\sinh x = -4$:

$$\Rightarrow x = \operatorname{arsinh}(-4) = \ln(-4 + \sqrt{(-4)^2 + 1}) = \ln(-4 + \sqrt{17})$$

If $\sinh x = 3$:

$$\Rightarrow x = \operatorname{arsinh} 3 = \ln(3 + \sqrt{3^2 + 1}) = \ln(3 + \sqrt{10})$$

$\therefore x = \ln(-4 + \sqrt{17})$ or $x = \ln(3 + \sqrt{10})$.

11 b)

$2\cosh^2 x - \cosh x - 21 = 0$

$$\Rightarrow (2\cosh x - 7)(\cosh x + 3) = 0 \Rightarrow \cosh x = \frac{7}{2} \text{ or } \cosh x = -3$$

$\cosh x = -3$ has no real solutions.

$\cosh x = \frac{7}{2}$:

$$\Rightarrow x = \pm \operatorname{arcosh} \frac{7}{2} = \pm \ln(\frac{7}{2} + \sqrt{(\frac{7}{2})^2 - 1})$$

$$\Rightarrow x = \pm\ln(\frac{7}{2} + \sqrt{\frac{45}{4}}) = \pm\ln(\frac{7+3\sqrt{5}}{2})$$

$\therefore x = \pm\ln(\frac{7+3\sqrt{5}}{2})$.

11 c)

$12\tanh^2 x + \tanh x - 1 = 0$

$$\Rightarrow (4\tanh x - 1)(3\tanh x + 1) = 0 \Rightarrow \tanh x = \frac{1}{4} \text{ or } \tanh x = -\frac{1}{3}$$

If $\tanh x = \frac{1}{4}$:

$$\Rightarrow x = \operatorname{artanh}\frac{1}{4} = \frac{1}{2}\ln(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}) = \frac{1}{2}\ln\frac{5}{3}$$

If $\tanh x = -\frac{1}{3}$:

$$\Rightarrow x = \operatorname{artanh}(-\frac{1}{3}) = \frac{1}{2}\ln(\frac{1+(-\frac{1}{3})}{1-(-\frac{1}{3})}) = \frac{1}{2}\ln\frac{1}{2} = -\frac{1}{2}\ln 2$$

$\therefore x = \frac{1}{2}\ln\frac{5}{3}$ or $x = -\frac{1}{2}\ln 2$.

12)

$3\cosh^2 x + 10\cosh x - 25 = 0$

$$\Rightarrow (3\cosh x - 5)(\cosh x + 5) = 0 \Rightarrow \cosh x = \frac{5}{3} \text{ or } \cosh x = -5$$

$\cosh x = -5$ has no real solutions.

$\cosh x = \frac{5}{3}$:

$$\Rightarrow x = \pm\operatorname{arcosh}\frac{5}{3} = \pm\ln(\frac{5}{3} + \sqrt{(\frac{5}{3})^2 - 1}) = \pm\ln(\frac{5}{3} + \frac{4}{3}) = \pm\ln 3$$

So $x = \ln 3$ or $x = -\ln 3 = \ln\frac{1}{3}$.

13)

$3\cosh^2 x - 17\cosh x - 56 = 0$

$$\Rightarrow (3\cosh x + 7)(\cosh x - 8) = 0 \Rightarrow \cosh x = 8 \text{ or } \cosh x = -\frac{7}{3}$$

$\cosh x = -\frac{7}{3}$ has no real solutions.

$\cosh x = 8$:

$$\Rightarrow x = \pm\operatorname{arcosh} 8 = \pm\ln(8 + \sqrt{8^2 - 1}) = \pm\ln(8 + 3\sqrt{7})$$

So $x = \ln(8+3\sqrt{7})$ or $x = -\ln(8+3\sqrt{7}) = \ln(\frac{1}{8+3\sqrt{7}}) = \ln(8-3\sqrt{7})$. $\therefore x = \ln(8+3\sqrt{7})$ or $x = \ln(8-3\sqrt{7})$.

14 a)

$\operatorname{arsinh} x = \ln 2$

$$\Rightarrow \ln(x + \sqrt{x^2+1}) = \ln 2 \Rightarrow x + \sqrt{x^2+1} = 2$$
$$\Rightarrow \sqrt{x^2+1} = 2 - x \Rightarrow x^2 + 1 = (2-x)^2$$
$$\Rightarrow x^2 + 1 = x^2 - 4x + 4 \Rightarrow 1 = -4x + 4 \Rightarrow 4x = 3 \Rightarrow x = \frac{3}{4}$$

Alternative Method:

$$\operatorname{arsinh} x = \ln 2 \Rightarrow x = \sinh(\ln 2) = \frac{e^{\ln 2} - e^{-\ln 2}}{2}$$
$$= \frac{e^{\ln 2} - e^{\ln(1/2)}}{2} = \frac{2 - \frac{1}{2}}{2} = \frac{3}{4}$$

14 b)

$\operatorname{arcosh} x = \ln 3$

$$\Rightarrow \ln(x + \sqrt{x^2 - 1}) = \ln 3 \Rightarrow x + \sqrt{x^2 - 1} = 3$$

$$\Rightarrow \sqrt{x^2 - 1} = 3 - x \Rightarrow x^2 - 1 = (3 - x)^2$$

$$\Rightarrow x^2 - 1 = x^2 - 6x + 9 \Rightarrow -1 = -6x + 9 \Rightarrow 6x = 10 \Rightarrow x = \frac{5}{3}$$

Alternative Method:

$$\operatorname{arcosh} x = \ln 3 \Rightarrow x = \cosh(\ln 3) = \frac{e^{\ln 3} + e^{-\ln 3}}{2}$$

$$= \frac{e^{\ln 3} + e^{\ln(1/3)}}{2} = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}$$

14 c)

$\operatorname{artanh} x = \ln 4$

$$\Rightarrow \frac{1}{2}\ln(\frac{1+x}{1-x}) = \ln 4 \Rightarrow \ln(\frac{1+x}{1-x}) = 2\ln 4$$

$$\Rightarrow \ln(\frac{1+x}{1-x}) = \ln 16 \Rightarrow \frac{1+x}{1-x} = 16$$

$$\Rightarrow 1 + x = 16 - 16x \Rightarrow 17x = 15 \Rightarrow x = \frac{15}{17}$$

Alternative Method:

$$\operatorname{artanh} x = \ln 4 \Rightarrow x = \tanh(\ln 4) = \frac{e^{2\ln 4} - 1}{e^{2\ln 4} + 1}$$

$$= \frac{e^{\ln 16} - 1}{e^{\ln 16} + 1} = \frac{16 - 1}{16 + 1} = \frac{15}{17}$$

14 d)

$\operatorname{arsinh} 2x = 2\ln 3$

$$\Rightarrow \operatorname{arsinh} 2x = \ln 9 \Rightarrow \ln(2x + \sqrt{(2x)^2 + 1}) = \ln 9$$

$$\Rightarrow 2x + \sqrt{4x^2 + 1} = 9 \Rightarrow \sqrt{4x^2 + 1} = 9 - 2x$$

$$\Rightarrow 4x^2 + 1 = (9 - 2x)^2 \Rightarrow 4x^2 + 1 = 4x^2 - 36x + 81$$

$$\Rightarrow 1 = -36x + 81 \Rightarrow 36x = 80 \Rightarrow x = \frac{20}{9}$$

Alternative Method:

$$\operatorname{arsinh} 2x = 2\ln 3 \Rightarrow \operatorname{arsinh} 2x = \ln 9 \Rightarrow 2x = \sinh(\ln 9)$$

$$\Rightarrow x = \frac{1}{2} \times \frac{e^{\ln 9} - e^{-\ln 9}}{2} = \frac{e^{\ln 9} - e^{\ln(1/9)}}{4}$$

$$= \frac{9 - \frac{1}{9}}{4} = \frac{20}{9}$$

14 e)

$\operatorname{arcosh} 3x = 3\ln 2$

$$\Rightarrow \operatorname{arcosh} 3x = \ln 8 \Rightarrow \ln(3x + \sqrt{(3x)^2 - 1}) = \ln 8$$

$$\Rightarrow 3x + \sqrt{9x^2 - 1} = 8 \Rightarrow \sqrt{9x^2 - 1} = 8 - 3x$$

$$\Rightarrow 9x^2 - 1 = (8 - 3x)^2 \Rightarrow 9x^2 - 1 = 64 - 48x + 9x^2$$

$$\Rightarrow -1 = 64 - 48x \Rightarrow 48x = 65 \Rightarrow x = \frac{65}{48}$$

Alternative Method:

$$\operatorname{arcosh} 3x = 3\ln 2 \Rightarrow \operatorname{arcosh} 3x = \ln 8 \Rightarrow 3x = \cosh(\ln 8)$$

$$\Rightarrow x = \frac{1}{3} \times \frac{e^{\ln 8} + e^{-\ln 8}}{2} = \frac{e^{\ln 8} + e^{\ln(1/8)}}{6}$$

$$= \frac{8 + \frac{1}{8}}{6} = \frac{65}{48}$$

14 f)

$\operatorname{artanh} \frac{1}{2}x = 2\ln \frac{1}{2}$

$$\Rightarrow \operatorname{artanh} \frac{1}{2}x = \ln \frac{1}{4} \Rightarrow \frac{1}{2}\ln(\frac{1 + \frac{1}{2}x}{1 - \frac{1}{2}x}) = \ln \frac{1}{4}$$

$$\Rightarrow \ln(\frac{1 + \frac{1}{2}x}{1 - \frac{1}{2}x}) = 2\ln \frac{1}{4} = \ln \frac{1}{16}$$

$$\Rightarrow \frac{1 + \frac{1}{2}x}{1 - \frac{1}{2}x} = \frac{1}{16} \Rightarrow 16(1 + \frac{1}{2}x) = 1 - \frac{1}{2}x$$

$$\Rightarrow 16 + 8x = 1 - \frac{1}{2}x \Rightarrow \frac{17}{2}x = -15 \Rightarrow x = -\frac{30}{17}$$

Alternative Method:

$$\operatorname{artanh} \frac{1}{2}x = 2\ln \frac{1}{2} \Rightarrow \operatorname{artanh} \frac{1}{2}x = \ln \frac{1}{4} \Rightarrow \frac{1}{2}x = \tanh(\ln \frac{1}{4})$$

$$\Rightarrow x = 2 \times \frac{e^{2\ln \frac{1}{4}} - 1}{e^{2\ln \frac{1}{4}} + 1} = 2 \times \frac{e^{\ln \frac{1}{16}} - 1}{e^{\ln \frac{1}{16}} + 1}$$

$$= 2 \times \frac{\frac{1}{16} - 1}{\frac{1}{16} + 1} = 2 \times (-\frac{15}{17}) = -\frac{30}{17}$$

15 a)

$x = \sinh(\ln(3 - x))$

$$\Rightarrow \operatorname{arsinh} x = \ln(3 - x) \Rightarrow \ln(x + \sqrt{x^2 + 1}) = \ln(3 - x)$$

$$\Rightarrow x + \sqrt{x^2 + 1} = 3 - x \Rightarrow \sqrt{x^2 + 1} = 3 - 2x$$

$$\Rightarrow x^2 + 1 = (3 - 2x)^2 \Rightarrow x^2 + 1 = 9 - 12x + 4x^2$$

$$\Rightarrow 0 = 3x^2 - 12x + 8 \Rightarrow 0 = 3(x^2 - 4x) + 8$$

$$\Rightarrow 0 = 3((x - 2)^2 - 4) + 8 \Rightarrow 0 = 3(x - 2)^2 - 12 + 8$$

$$\Rightarrow 3(x-2)^2 = 4 \Rightarrow (x-2)^2 = \frac{4}{3} \Rightarrow x-2 = \pm\frac{2\sqrt{3}}{3}$$

$$\Rightarrow x = 2 \pm \frac{2\sqrt{3}}{3}$$

$x = 2 - \frac{2\sqrt{3}}{3}$ only as $\ln(3-x)$ is undefined when $x = 2 + \frac{2\sqrt{3}}{3}$. Alternative Method:

$$x = \sinh(\ln(3-x)) \Rightarrow x = \frac{e^{\ln(3-x)} - e^{-\ln(3-x)}}{2}$$

$$\Rightarrow 2x = e^{\ln(3-x)} - e^{\ln((3-x)^{-1})} \Rightarrow 2x = 3 - x - \frac{1}{3-x}$$

$$\Rightarrow \frac{1}{3-x} = 3 - 3x \Rightarrow 1 = (3-x)(3-3x)$$

$$\Rightarrow 1 = 9 - 12x + 3x^2 \Rightarrow 0 = 3x^2 - 12x + 8$$

$$\Rightarrow 0 = 3(x^2 - 4x) + 8 \Rightarrow 0 = 3((x-2)^2 - 4) + 8$$

$$\Rightarrow 0 = 3(x-2)^2 - 12 + 8 \Rightarrow 3(x-2)^2 = 4 \Rightarrow (x-2)^2 = \frac{4}{3}$$

$$\Rightarrow x - 2 = \pm\frac{2\sqrt{3}}{3} \Rightarrow x = 2 \pm \frac{2\sqrt{3}}{3}$$

$x = 2 - \frac{2\sqrt{3}}{3}$ only as $\ln(3-x)$ is undefined when $x = 2 + \frac{2\sqrt{3}}{3}$.

15 b)

$\cosh(\ln(2x-1)) = x + 2$

$$\Rightarrow \ln(2x-1) = \operatorname{arcosh}(x+2) = \ln(x + 2 + \sqrt{(x+2)^2 - 1})$$

$$\Rightarrow 2x - 1 = x + 2 + \sqrt{x^2 + 4x + 3} \Rightarrow x - 3 = \sqrt{x^2 + 4x + 3}$$

$$\Rightarrow (x-3)^2 = x^2 + 4x + 3 \Rightarrow x^2 - 6x + 9 = x^2 + 4x + 3$$

$$\Rightarrow -6x + 9 = 4x + 3 \Rightarrow 6 = 10x \Rightarrow x = \frac{3}{5}$$

Alternative Method:

$$\cosh(\ln(2x-1)) = x + 2 \Rightarrow \frac{e^{\ln(2x-1)} + e^{-\ln(2x-1)}}{2} = x + 2$$

$$\Rightarrow e^{\ln(2x-1)} + e^{\ln((2x-1)^{-1})} = 2x + 4 \Rightarrow 2x - 1 + \frac{1}{2x-1} = 2x + 4$$

$$\Rightarrow \frac{1}{2x-1} = 5 \Rightarrow 2x - 1 = \frac{1}{5} \Rightarrow x = \frac{3}{5}$$

15 c)

$x - 1 = \tanh(\ln\sqrt{x+1})$

$$\Rightarrow \operatorname{artanh}(x-1) = \ln\sqrt{x+1} \Rightarrow \frac{1}{2}\ln(\frac{1+(x-1)}{1-(x-1)}) = \frac{1}{2}\ln(x+1)$$

$$\Rightarrow \ln(\frac{x}{2-x}) = \ln(x+1) \Rightarrow \frac{x}{2-x} = x + 1$$

$$\Rightarrow x = (x+1)(2-x) \Rightarrow x = -x^2 + x + 2 \Rightarrow x^2 = 2$$

$x = \sqrt{2}$ only as $\sqrt{x+1}$ is not real when $x = -\sqrt{2}$. Alternative Method:

$$x - 1 = \tanh(\ln\sqrt{x+1}) \Rightarrow x - 1 = \frac{e^{2\ln\sqrt{x+1}} - 1}{e^{2\ln\sqrt{x+1}} + 1}$$

$$\Rightarrow x - 1 = \frac{e^{\ln(x+1)} - 1}{e^{\ln(x+1)} + 1} \Rightarrow x - 1 = \frac{x+1-1}{x+1+1} = \frac{x}{x+2}$$

$$\Rightarrow (x-1)(x+2) = x \Rightarrow x^2 + x - 2 = x \Rightarrow x^2 = 2$$

$x = \sqrt{2}$ only as $\sqrt{x+1}$ is not real when $x = -\sqrt{2}$.

16 a)

$\operatorname{artanh} x - \ln(1+x) = \ln\frac{6}{5}$

$$\frac{1}{2}\ln(\frac{1+x}{1-x}) - \ln(1+x) = \ln\frac{6}{5}$$

$$\Rightarrow \ln(\frac{1+x}{1-x}) - 2\ln(1+x) = 2\ln\frac{6}{5}$$

$$\Rightarrow \ln(\frac{1+x}{1-x}) - \ln(1+x)^2 = \ln\frac{36}{25}$$

$$\Rightarrow \ln(\frac{1+x}{1-x} \times \frac{1}{(1+x)^2}) = \ln\frac{36}{25} \Rightarrow \frac{1}{(1-x)(1+x)} = \frac{36}{25}$$

$$\Rightarrow (1-x)(1+x) = \frac{25}{36} \Rightarrow 1 - x^2 = \frac{25}{36} \Rightarrow x^2 = \frac{11}{36} \Rightarrow x = \pm\frac{\sqrt{11}}{6}$$

16 b)

$\ln 2 + \ln(1+x) - \operatorname{artanh} x = \ln\frac{2}{9}$

$$\Rightarrow \ln(2(1+x)) - \frac{1}{2}\ln(\frac{1+x}{1-x}) = \ln\frac{2}{9}$$

$$\Rightarrow 2\ln(2(1+x)) - \ln(\frac{1+x}{1-x}) = 2\ln\frac{2}{9}$$

$$\Rightarrow \ln(4(1+x)^2) - \ln(\frac{1+x}{1-x}) = \ln\frac{4}{81}$$

$$\Rightarrow \ln(4(1+x)^2 \times \frac{1-x}{1+x}) = \ln\frac{4}{81} \Rightarrow 4(1+x)(1-x) = \frac{4}{81}$$

$$\Rightarrow 1 - x^2 = \frac{1}{81} \Rightarrow x^2 = \frac{80}{81} \Rightarrow x = \pm\frac{4\sqrt{5}}{9}$$

16 c)

$\operatorname{arsinh} x - \ln(1+x) = \ln 2$

$$\Rightarrow \ln(x + \sqrt{x^2+1}) - \ln(1+x) = \ln 2$$

$$\Rightarrow \ln(x + \sqrt{x^2+1}) = \ln 2 + \ln(1+x) = \ln(2+2x)$$

$$\Rightarrow x + \sqrt{x^2+1} = 2 + 2x \Rightarrow \sqrt{x^2+1} = 2 + x$$

$$\Rightarrow x^2 + 1 = (2+x)^2 \Rightarrow x^2 + 1 = 4 + 4x + x^2$$

$$\Rightarrow 1 = 4 + 4x \Rightarrow 4x = -3 \Rightarrow x = -\frac{3}{4}$$

16 d)

$\ln 3 + \ln(x-1) - \operatorname{arcosh} x = 0$

$$\Rightarrow \ln(3x-3) - \ln(x+\sqrt{x^2-1}) = 0 \Rightarrow \ln(3x-3) = \ln(x+\sqrt{x^2-1})$$
$$\Rightarrow 3x-3 = x+\sqrt{x^2-1} \Rightarrow 2x-3 = \sqrt{x^2-1}$$
$$\Rightarrow (2x-3)^2 = x^2-1 \Rightarrow 4x^2-12x+9 = x^2-1$$
$$\Rightarrow 3x^2-12x+10 = 0 \Rightarrow 3(x^2-4x)+10 = 0$$
$$\Rightarrow 3((x-2)^2-4)+10 = 0 \Rightarrow 3(x-2)^2-12+10 = 0$$
$$\Rightarrow (x-2)^2 = \frac{2}{3} \Rightarrow x-2 = \pm\sqrt{\frac{2}{3}} \Rightarrow x = 2 \pm \sqrt{\frac{2}{3}}$$

$x = 2+\sqrt{\frac{2}{3}}$ only as $x = 2-\sqrt{\frac{2}{3}}$ does not solve the original equation.

17 a)

$\operatorname{arsinh} x + \ln 2 = \operatorname{arcosh} x + \ln 3$

$$\Rightarrow \ln(x+\sqrt{x^2+1}) + \ln 2 = \ln(x+\sqrt{x^2-1}) + \ln 3$$
$$\Rightarrow \ln(2x+2\sqrt{x^2+1}) = \ln(3x+3\sqrt{x^2-1})$$
$$\Rightarrow 2x+2\sqrt{x^2+1} = 3x+3\sqrt{x^2-1}$$
$$\Rightarrow 2\sqrt{x^2+1}-3\sqrt{x^2-1} = x$$
$$\Rightarrow (2\sqrt{x^2+1}-3\sqrt{x^2-1})^2 = x^2$$
$$\Rightarrow 4(x^2+1)-12\sqrt{x^4-1}+9(x^2-1) = x^2$$
$$\Rightarrow 4x^2+4+9x^2-9-x^2 = 12\sqrt{x^4-1} \Rightarrow 12x^2-5 = 12\sqrt{x^4-1}$$
$$\Rightarrow (12x^2-5)^2 = 144(x^4-1) \Rightarrow 144x^4-120x^2+25 = 144x^4-144$$
$$\Rightarrow -120x^2+25 = -144 \Rightarrow 120x^2 = 169 \Rightarrow x^2 = \frac{169}{120} \Rightarrow x = \pm\frac{13\sqrt{30}}{60}$$

$x = \frac{13\sqrt{30}}{60}$ only as $x \geq 1$ for $\operatorname{arcosh} x$.

17 b)

$\operatorname{arsinh} x - 2\operatorname{arcosh} x = 0$

$$\Rightarrow \ln(x+\sqrt{x^2+1}) - 2\ln(x+\sqrt{x^2-1}) = 0$$
$$\Rightarrow \ln(x+\sqrt{x^2+1}) = \ln(x+\sqrt{x^2-1})^2$$
$$\Rightarrow x+\sqrt{x^2+1} = (x+\sqrt{x^2-1})^2$$
$$\Rightarrow x+\sqrt{x^2+1} = x^2+2x\sqrt{x^2-1}+x^2-1$$
$$\Rightarrow \sqrt{x^2+1}-2x\sqrt{x^2-1} = 2x^2-x-1$$
$$\Rightarrow (\sqrt{x^2+1}-2x\sqrt{x^2-1})^2 = (2x^2-x-1)^2$$
$$\Rightarrow x^2+1-4x\sqrt{x^4-1}+4x^2(x^2-1) = 4x^4-4x^3-3x^2+2x+1$$
$$\Rightarrow 4x^3-2x = 4x\sqrt{x^4-1} \Rightarrow 2x^2-1 = 2\sqrt{x^4-1} \text{ as } x \neq 0$$
$$\Rightarrow (2x^2-1)^2 = 4(x^4-1) \Rightarrow 4x^4-4x^2+1 = 4x^4-4$$
$$\Rightarrow -4x^2+1 = -4 \Rightarrow 4x^2 = 5 \Rightarrow x^2 = \frac{5}{4} \Rightarrow x = \pm\frac{\sqrt{5}}{2}$$

$x = \frac{\sqrt{5}}{2}$ only as $x \geq 1$ for $\operatorname{arcosh} x$.